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Peach Springs Tuff: Its Bearing on Structural Evolution of the Colorado Plateau and Development of Cenozoic Drainage in Mohave County, Arizona

ABSTRACT

The Peach Springs Tuff, a welded ash flow of Miocene age, formerly blanketed a minimum area of 2,000 sq mi on both sides of the western edge of the Colorado Plateau in Mohave County, Arizona, and filled northeast-trending, pre–Colorado River canyons cut through the Paleozoic rocks.

Paleomagnetic direction, petrography, and field relations indicate that the tuff is a single cooling unit of trachytic composition and normal polarity. It was erupted across the region prior to the last major movement of the plateau marginal faults, the completion of a significant amount of Basin and Range block faulting, and development of the existing topographic relief. A pre-tuff period of uplift and erosion exposed the Precambrian basement rocks directly west of the plateau.

Gravel buried in canyons beneath the tuff on the edge of the Hualapai Plateau is evidence of northeast-flowing, incised drainage, in pre-tuff time, which probably continued northward along the Hurricane fault zone across the region where the younger canyon of the Colorado River is now. This ancient drainage system was disrupted by faulting, volcanism, and localized fluvial aggradation, followed by gradual incision of the modern Colorado River drainage.

South of the Truxton Valley, there is evidence of a pre–middle Miocene drainage divide along the edge of the present plateau.

All geologic evidence points to an early to middle Pliocene age for the modern Colorado River and associated tributary drainage on the Hualapai Plateau west of the Hurricane fault.

INTRODUCTION

In this paper, the eruptive nature and widespread distribution of the Peach Springs Tuff and associated Cenozoic rocks are related to the structural and erosional history of the Basin and Range–Colorado Plateau boundary and to events leading to the development of the Grand Canyon of the Colorado River. Investigations of the Peach Springs Tuff developed from a more general study of the Cenozoic geology of the Hualapai Plateau done by R. A. Young from 1962 through 1966. During the spring of 1970 and the summer of 1971, W. J. Brennan joined the field studies to collect samples for plageomagnetic measurements and to make petrologic analyses of the tuff.

PREVIOUS GEOLOGIC INVESTIGATIONS

Geologic reconnaissance in Mohave County was done by Marvine (1875) and Lee (1908). More detailed studies of portions of the area were carried out by Thomas (1953), Gray (1959), and Peirce (1972). The reports by Twenter (1962) and Gillespie and Bentley (1971) both contain geologic base maps (1:125,000) which are useful for examining in some detail the distribution of rock types discussed in this paper.

Detailed maps of the Cenozoic geology of the Hualapai Plateau and the Truxton Valley have been compiled from aerial photographs (1:31,680) by Young (1966).

REGIONAL CENOZOIC GEOLOGY

Geologic evidence for the structural evolution of the Colorado Plateau in Arizona and the origin of the Grand Canyon of the Colorado River has been accumulating for the past ten years and has been summarized by McKee and others (1967), Hunt (1969), Lucchitta (1972), and others. There is ample evidence, supported by numerous radiometric dates, of uplift along the southern and southwestern margins of the plateau in Arizona during the Miocene and Pliocene Epochs. The uplift caused erosion of the Precambrian basement rocks and resulted in deposition of arkosic gravel onto the Paleozoic rocks of the plateau to the north and northeast. The gravel deposits are generally preserved under Tertiary lava flows in places on the edges of the plateau. Recent radiometric ages of volcanic clasts reported by McKee and McKee (1972) demonstrate that some of the north-flowing drainage persisted until early to middle Pliocene time in the area immediately east of the Hualapai Plateau (Fig. 1). It has also been substantiated that the modern Colorado River did not exist to the west of the Hurricane fault until Pliocene time, as discussed most recently in the summary by Lucchitta (1972).

Cenozoic deposits fill Hualapai Plateau canyons, which converge on the Hurricane fault zone in Peach Springs Canyon, and these deposits document the existence of valleys 1 to 3 mi wide; the valleys eroded 1,500 ft into the plateau surface, within 8 mi of the Colorado River gorge (Young, 1966, 1970). The excavation of these large canyons to such depths must have required a major outlet to the north(?), which would account for the northerly flow measurements of imbrication in the basal arkosic gravel and the drainage trends indicated by the distribution of similar gravel derived from central Arizona, at higher elevations

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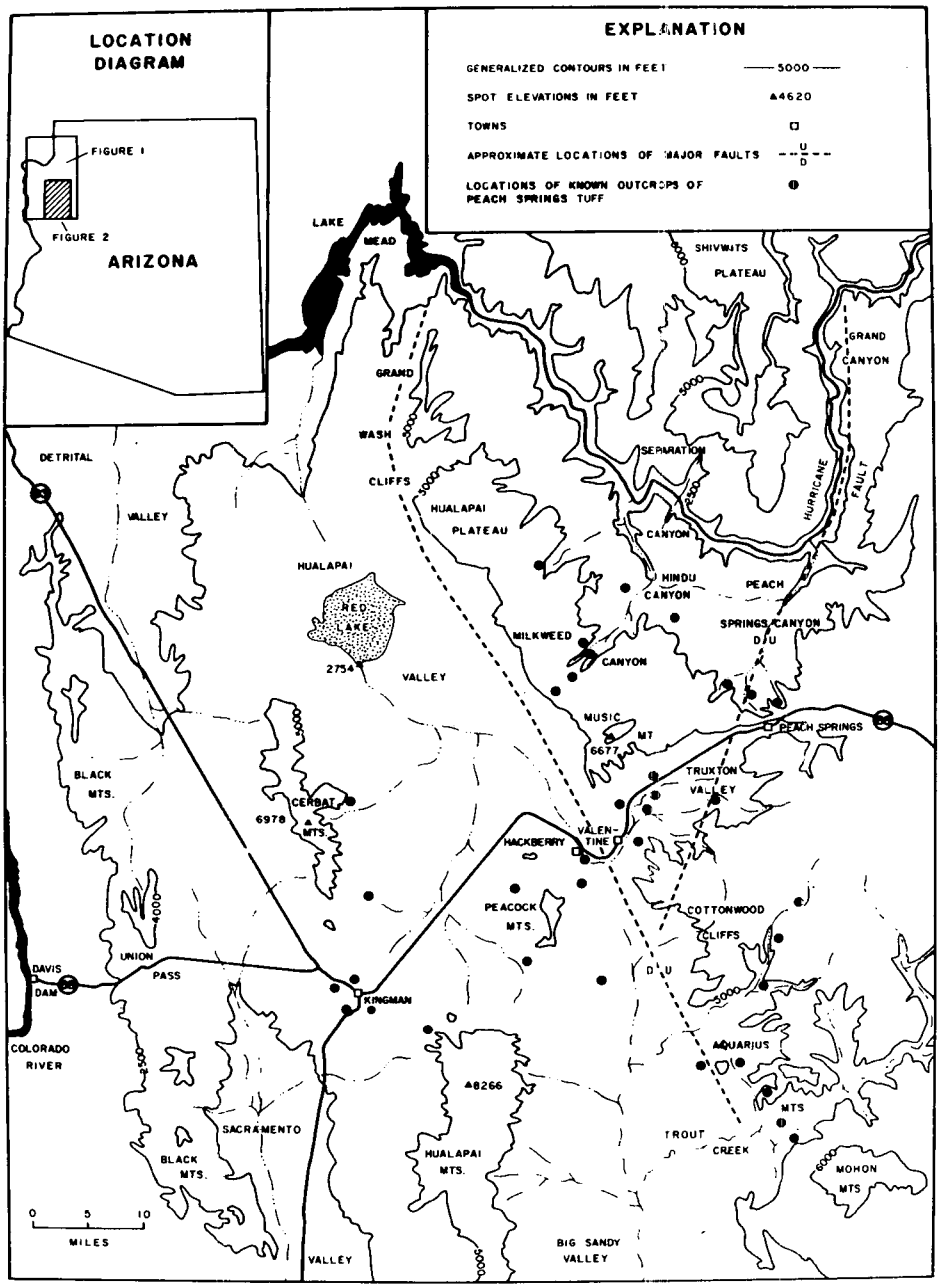


Figure 1. Generalized locations of known outcrops of Peach Springs Tuff and localities referred to in text.

east of the Hurricane fault on the Coconino Plateau (Young, 1966, 1970; McKee and McKee, 1972). The outlet for this major pre–middle Miocene drainage system on the Hualapai Plateau could not have been through the western Grand Canyon region to the Lake Mead area; the facies distribution and Pliocene age of the Muddy Creek sediments and volcanic rocks through which the Colorado River has eroded its channel preclude this (Longwell, 1936; Lucchitta, 1966, 1972). Nor could any hypothetical ancestral Colorado River have flowed southward through Peach Springs Canyon prior to discharging into the Grand Wash Trough (Lake Mead), as suggested by Hunt (1969, p. 120, 121). Such a southwesterly course is contrary to much of the evidence presented by Lucchitta (1972) and in this paper concerning the regional northeast slope, the structural evolution of the region, and the paleodrainage pattern. Furthermore, if the 200 ft of visible post-tuff faulting on the Hurricane fault in Peach Springs Canyon is removed, along with an equal amount determined from well data near Truxton, Arizona, the floor of the buried channel through the Truxton Valley into Peach Springs Canyon most logically would have sloped to the northeast prior to faulting. This reconstruction is not precise because the amount of regional tilting or warping which could have accompanied the faulting is unknown.

All major pre–middle Miocene drainage channels on the Hualapai Plateau and south to Trout Creek were blocked by the eruption of the Peach Springs Tuff, a welded ash flow which predates the Colorado River tributaries on the Hualapai Plateau. North-flowing, pre–Colorado River drainage north of the modern Colorado River along the Hurricane fault zone would have to have crossed Paleozoic rocks that are now topographically and stratigraphically 1,500 ft higher than the basal Cenozoic gravels in Peach Springs Canyon. Such northerly drainage could have resulted only from greater northeastward tilting of the edge of the Colorado Plateau during the early stages of the marginal uplift which preceded or accompanied the Basin and Range block faulting. Evidence for such a sequence of events is provided by the elevations and distribution of the Peach Springs Tuff outcrops and older Cenozoic deposits.

The Peach Springs Tuff is significant to an understanding of middle to late Tertiary events along this important boundary between two major geologic provinces because of (1) its widespread distribution in an area where Cenozoic rocks are difficult to correlate; (2) its relatively uniform, thin character across a region of significant structural and erosional relief; (3) the evidence it shows of faulting and regional uplift since middle Miocene time; and (4) the nature of the pre-tuff deposits and topography in buried canyons along the edge of the Colorado Plateau.

AGE AND DISTRIBUTION OF THE PEACH SPRINGS TUFF

Peach Springs Tuff specimens collected near Kingman and in Milkweed Canyon, Arizona, have been dated as 16.9 ± 0.4 m.y. (Damon and others, 1966, p. 28) and as 18.3 ± 0.6 m.y. (Damon and others, 1964, p. 19), respectively. The descrepancy between these two ages caused Damon to doubt the correlation of the two outcrops (Damon and others, 1966, p. 28); but the evidence we present substantiates the single-cooling-unit character of the tuff throughout the study area.

The distribution and thicknesses of known outcrops of the Peach Springs Tuff are plotted on Figure 2. The trend of outcrop thicknesses indicates that the source of the deposits was somewhere west of the Cerbat Mountains, most likely in or near the Black Mountains.

Figure 2 shows that the easternmost outcrops, critical for interpretation of the

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structural evolution of the Colorado Plateau, are very similar in thickness because of their nearly uniform distance from the center of the eruption. The thickness and complexity of the unit increase gradually from the Hualapai Plateau westward to the vicinity of Kingman. The easternmost exposures record deposition of a basal ash (distinctly bedded), averaging nearly 1 ft thick, followed by deposition of nearly 10 ft of loosely welded, nonlaminated, normally pinkish (variable) tuff which in turn is overlain by a densely welded ash flow 30 to 60 ft thick. This upper zone tends to be dark purple or gray, and in places has eroded back in a series of benches, which do not appear to mark any visible lithologic change. The densely welded upper portion of the tuff becomes noticeably less welded along its eastern edge, where it contains large fragments of pumice, cinders, and small cavities where similar fragments have been weathered out.

In contrast, the westernmost exposures (Fig. 2) that were identified as Peach Springs Tuff, northwest of Kingman, exhibit a basal ash layer 30 in. thick, a 137-ft-thick middle zone of loosely welded tuff containing a significant number of volcanic pebbles of various lithologies (xenoliths), and an upper, densely welded zone 84 ft thick. Similar types of pyroclastic deposits are present near Union Pass, where Route 68 crosses outcrops mapped by Longwell (1963) as Patsy Mine and Golden Door Volcanics (revised by Anderson, 1971, p. 45). Some radiometric ages for a welded ash-flow tuff in the Patsy Mine Volcanics closely match the ages determined for the Peach Springs Tuff (Anderson and others, 1972, p. 278). However, limited reconnaissance in the area did not enable us to visually correlate the Peach Springs Tuff with any specific portion of the thick section of pyroclastic rocks visible near Union Pass in the Black Mountains. The gradual change in thickness and degree of welding of the different zones within the ash flow between Peach Springs and Kingman suggest there would be an increased thickening and perhaps a more complex stratification toward the source area that would prevent a simple correlation with tuffs in the Black Mountains, where several similar volcanic units are involved in major faulting and in some areas have been slightly metamorphosed.

The basal ash was exposed and examined at 20 outcrops. A simple color zonation and grain-size variation of the ash laminae can be traced between some adjacent outcrops. The ash is distinctly bedded and exhibits a gradual easterly decrease in thickness (Fig. 2). The southernmost outcrops near Trout Creek, where the ash was thinnest, show most clearly bedding features caused by

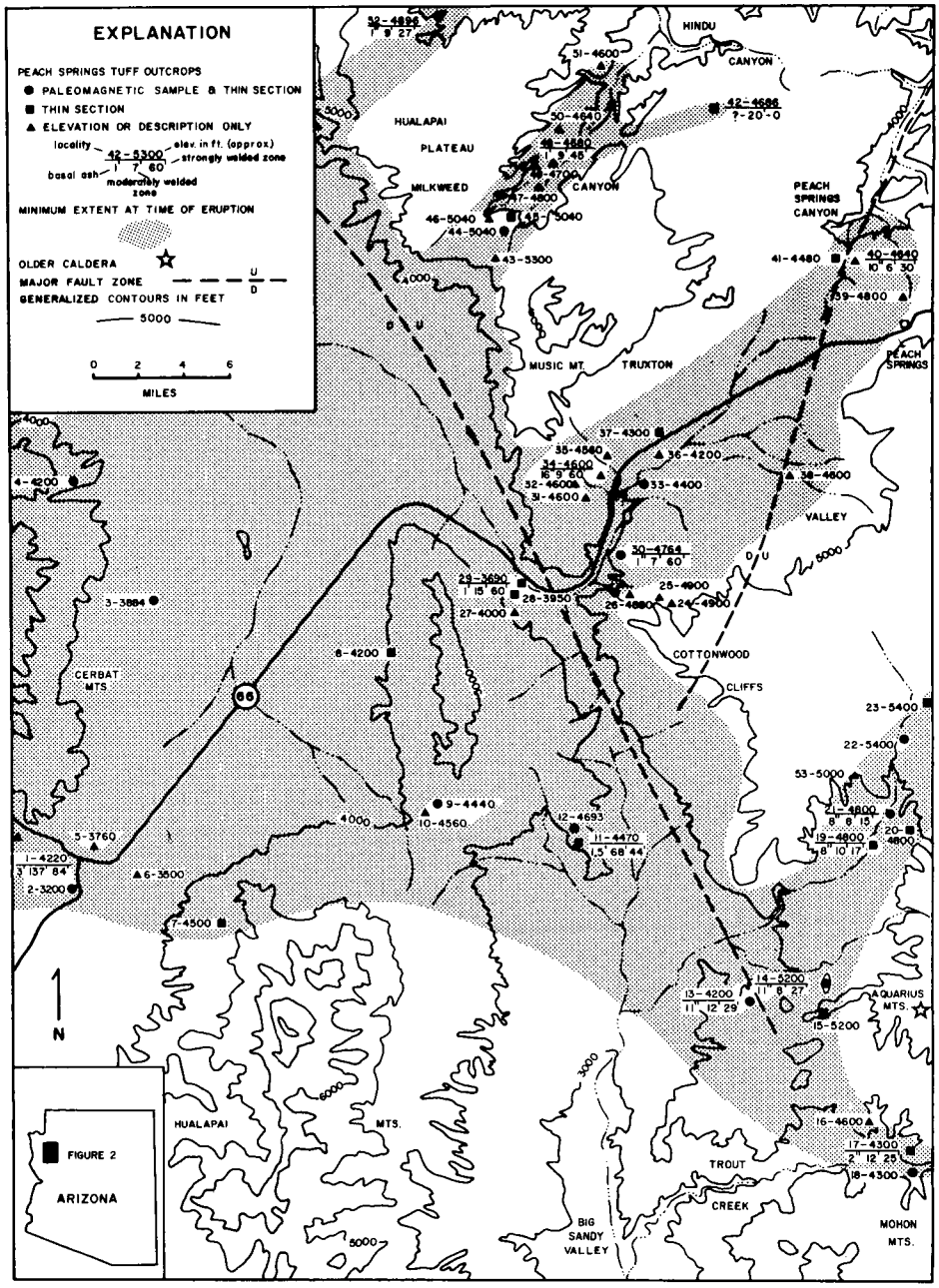


Figure 2. Minimum extent and thickness distribution of Peach Springs Tuff outcrops. Elevations from recently published and preliminary editions of U.S. Geol. Survey 1:24,000 topographic quadrangle maps.

minor surface irregularities and probable scouring during the deposition of the ash flow. Convincing evidence of base-surge structures or significant scour effects in the ash was observed only in the sides of the buried channel remnants beneath the tuff-capped mesas near Valentine, south of U.S. Route 66. At location 30 (Fig. 2), ~3 ft of whitish, nonlaminated ash mixed with cinders and locally included volcanic pebbles show barely discernible antidune(?) cross-bedding (Fisher and Waters, 1970, p. 161, 165) inclined toward the source (west) above a 1-in.-thick, discontinuous bed of basal ash.

The distribution of outcrops and elevations shown in Figure 2, combined with the regional thickness trends of the tuff and its uniformly varying characteristics, can best be explained by the following regional setting: The tuff erupted and flowed over a relatively featureless, gently sloping surface eroded down to the Precambrian basement but lacking the extreme relief of the modern Basin and Range province. Deflection of the ash flow between the Black Mountains and

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the present edge of the Colorado Plateau by north-south valleys or any significant, positive topographic features would have prevented the tuff from reaching its present position at the edge of the plateau or from flowing down canyons in the Paleozoic rocks along the edge of the plateau at localities 23, 40, and 51 (Fig. 2). If topographic features such as those which exist now had influenced the depositional pattern from the postulated source area west of Kingman, the ash flow would probably have filled only the Sacramento–Detrital Valley system; or if the ash flow had spilled into valleys similar to the modern Hualapai–Big Sandy Valley system, the thickness and distribution of the tuff would now be more indicative of restricted flow into such large valleys. Instead, the elevations, the wide distribution, and the gradual thickness changes over tens of miles suggest a gently sloping surface with few obstacles between Kingman and the Colorado Plateau.

The tuff is offset 1,000 ft along the major plateau marginal fault(s) near Hackberry and near the Aquarius Mountains. Displacement of 100 to 200 ft is visible on the Hurricane fault in Peach Springs Canyon and across the Truxton Valley. The tuff also has structural dips along the flanks of all the ranges associated with Basin and Range faulting, and is assumed to have partially or completely covered the area of the modern Cerbat and Peacock Mountains (Fig. 2).

PETROGRAPHY

The Peach Springs Tuff is an ash-flow sheet, as defined by Ross and Smith (1961, p. 3), which appears to have formed as the result of a single eruptive pulse. As such it is a single cooling unit.

In general, the Peach Springs Tuff contains only 4 to 14 percent phenocrysts. It is characterized by large (up to 4 mm) phenocrysts of sanidine, which commonly shows a pronounced chatoyancy. At many sampling localities, no other minerals are identifiable. The tuff's color ranges from grayish pink to light orange (5R 8/2 to 10R 6/6). At some outcrops, the uppermost portion is reddish purple (5RP 6/2 to 5RP 7/2).

In thin section, the groundmass of the welded zones is shard rich and contains few lithic fragments and little pumice, except in the uppermost portion. At some localities the shards are glassy, but at most localities they show evidence of devitrification, particularly, axiolitic structures within them. Phenocrysts range from euhedral to anhedral, and the quartz, sanidine, and plagioclase commonly show extensive resorption. The plagioclase is oligoclase ranging in composition from An_{14} to An_{33} and averaging An_{24}. Oscillatory compositional zoning is common in the plagioclase phenocrysts that are larger than ~2 mm diam.

TABLE 1. PHENOCRYST MINERALOGY OF PEACH SPRINGS TUFF

Quartz	1.0%
Sanidine	74.0%
Oligoclase	18.9%
Hornblende	2.0%
Biotite	0.7%
Clinopyroxene	0.1%
Sphene	0.8%
Opaque oxides	2.6%
Total	100.1%

Table 1 summarizes the petrography. The phenocryst values are averages from modal analyses of samples taken at localities indicated in Figure 2 and are representative of the mineralogy at all sampled localities.

The Peach Springs Tuff has been studied near Kingman, Arizona, by Thomas (1953), who classified it as a rhyolite and called it the Kingman rhyolite series. The phenocryst mineralogy determined in our study suggests that the tuff is, in fact, trachytic in composition (Wahlstrom, 1955; Huang, 1962; Petersen, 1961). However, O'Connor (1965) and others have shown that modal classifications are inadequate to completely describe aphanitic or glassy volcanic rocks. Lipman and others (1966) have pointed out that the groundmasses of porphyritic volcanic rocks tend to be more silicic than their bulk compositions. For this reason, the classification of the Peach Springs Tuff as a trachyte is tentative.

Opaque mineral grains range in size from 0.6 mm to less than 1 μ and are composed of titanomagnetite, which in some grains is extensively oxidized to hematite along irregular cracks and along (III) planes. Ilmenite is present but not abundant. Curie temperatures range from 565° to 585°C and indicate that low-titanium magnetite is the most abundant magnetic mineral phase, as petrographic observations suggest.

PALEOMAGNETIC METHODS

Oriented block samples were collected from 14 widely distributed outcrops of the Peach Springs Tuff (Fig. 2). An additional 14 small basaltic and andesitic flows, which occur both above and below the tuff on the Hualapai Plateau and in the Aquarius Mountains, were also sampled. The number of samples collected at each site varied from three to five. One drill core was taken from each sample.

Directions of remanent magnetization of all cores were measured on a Princeton Applied Research SM2 spinner magnetometer. The cores were magnetically cleaned in peak ac fields up to 500 oe; the optimum field at most sampling sites was 300 oe. A small number of cores whose magnetic directions did not attain a stable orientation during magnetic cleaning in progressively higher ac fields were rejected (McElhinny and Gough, 1963). The anomalously high NRM exhibited by these cores and the absence of any other observable differences indicate that the instability is due to local remagnetization by lightning.

PALEOMAGNETIC RESULTS

Equal-area projections of the magnetic directions in the Peach Springs Tuff and associated middle Miocene flows are shown in Figure 3. The average magnetic direction of all 15 flows was approximated by applying Fisher's method to the 15 mean directions (Irving, 1964; Fisher, 1953).

Although the Peach Springs Tuff is the only one of the 15 flows for which a radiometric age has been determined, field relationships suggest that all of the flows were erupted during a relatively short interval of time, probably only a few million years. Thus the average magnetic direction of all 15 flows is assumed to be representative of the average geomagnetic field direction for northwestern Arizona during a part of middle Miocene time. An approximate middle Miocene paleomagnetic pole position calculated from the average magnetic directions of the 15 flows is shown in Figure 4. Like most other paleomagnetic pole positions for Miocene rocks in western North America, it lies close to the rotational pole (Irving, 1964; Watkins, 1965; Symons, 1969; York and others, 1971). This agreement supports our assumption that the average magnetic direction of all 15 flows is representative of rocks of middle Miocene age in northwestern Arizona.

The scatter in the magnetic directions between the flows in the study area is due to the secular variation of the geomagnetic field (Irving, 1964), which Grommé and others (1972) have shown to be useful in the correlation of individual cooling units. Figure 3 illustrates that the magnetic declination of the Peach Springs Tuff is markedly different from the average direction, indicating that it was erupted at a time when the declination of the geomagnetic field was far to the east of its average value. This resulted in the fortuitous acquisition of a distinctive paleomagnetic direction by the tuff (declination 32.3°, inclination 42.8°, Fisher precision parameter -82.6, semiangle of cone of 95 percent confidence $\alpha 95 = 4.4°$). This confirms the petrologic and stratigraphic evidence that indicates that the Peach Springs Tuff is a single cooling unit.

CENOZOIC GEOLOGIC HISTORY

The canyons cut into Precambrian and Cambrian rocks at Milkweed, Hindu, and Peach Springs canyons (Figs. 1 and 5) are filled with extensive Tertiary deposits, mapped in detail by Young (1966, 1970).

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The basal arkosic gravel in these channels indicates extensive erosion of Precambrian crystalline terrain to the west by streams that were deeply incised and through-flowing on the plateau (Fig. 5). These canyons were blocked by structural movements which resulted in massive, locally derived fanglomerates and locally occurring lacustrine limestones interbedded with siltstone and claystone in Milkweed and Peach Springs Canyons. Following the establishment of new local drainage (on the plateau), the deep canyons were almost completely filled with gravel derived mainly from the Paleozoic limestones cropping out on the local divides on the Hualapai Plateau (Fig. 5) and around the Truxton Valley. The uppermost beds of the gravel immediately beneath the volcanic section including the Peach Springs Tuff contain increasing amounts of Precambrian crystalline rock types derived from local exposures on the southwest side of the Truxton Valley, from west of Hackberry, and from west of the plateau at the head of Milkweed Canyon. These crystalline clasts indicate partial or temporary re-establishment of drainage from the west across the modern plateau fault scarp. This episode was ended by the regional volcanism during which the Peach Springs Tuff erupted (middle Miocene).

Beneath the tuff are locally erupted basalts of limited extent interbedded with fluvially reworked volcanic sediments and agglomerates. The volcanic sediments are everywhere directly associated with the locally erupted volcanic flows and are generally thickest beneath them. Under the tuff, near the center of the Truxton Valley, is 50 ft of basalt, in a well which also penetrates 233 ft of postvolcanic gravel. The volcanism appears to have begun to the west of the plateau and proceeded in an eastward direction, as shown by the consistent position of the thin plateau flows and interbedded Peach Springs Tuff above gravel containing volcanic clasts derived from the west. Minor eruptive vents on the plateau were present, as shown by thin dikes and small plugs in gravel near the head of Milkweed Canyon.

Imbrication in fluviovolcanic sediments exposed beneath the lava at the very edge of the Hualapai Plateau in all of the buried canyons shows that streams were continuing to breach the plateau escarpment from the west, implying a regional slope from the southwest to the northeast. However, directional indicators in equivalent volcanic rocks and gravel immediately beneath the Peach Springs Tuff in channels south of the Truxton Valley along the Cottonwood Cliffs and Aquarius Mountains suggest the existence of a northwest-trending drainage divide parallel to the present margin of the Colorado Plateau prior to the eruption of the Peach Springs Tuff. Several tributaries

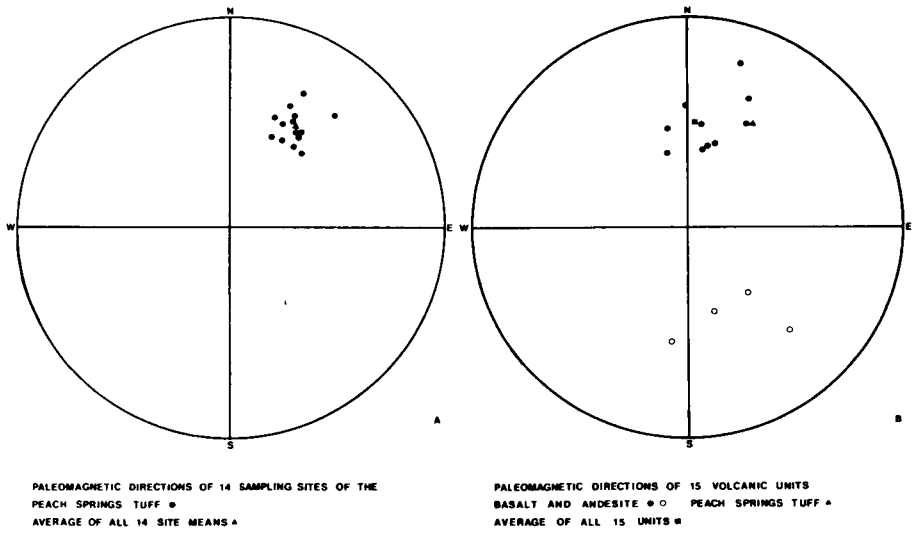


Figure 3. Equal-area projections of remanent magnetization directions of some Tertiary volcanic rocks in northwestern Arizona, after ac demagnetization. Open symbols are directions on the upper hemisphere; solid symbols are directions on the lower hemisphere.

preserved under the tuff between Valentine and Trout Creek contain gravel with distinct imbrication, showing drainage westward off the present plateau toward the Big Sandy Valley. Thus, there is evidence of uplift and erosion along the edge of the plateau in this area before eruption of the Peach Springs Tuff. Post-tuff displacement, along the margin of the plateau, is relatively down to the west, in apparent contradiction to the several thousand feet of uplift of the adjacent ranges to the west that has raised the Precambrian basement above the Paleozoic rocks on the plateau. This drainage off the plateau in pre-tuff time may have taken the form of small (minor) alluvial fans, similar to those present today along the plateau escarpment, which do not necessarily reflect the regional drainage trends of the major rivers.

EVOLUTION OF THE PLATEAU–BASIN AND RANGE BOUNDARY

We hypothesize a broad uplift west of the Colorado Plateau to form the erosion surface on which the tuff erupted and to provide the northeast slope for the major pre-tuff drainage. An unspecified amount of pre-tuff movement on faults along the plateau margins might have created a minor but significant topographic boundary breached by streams at a few points. This is substantiated by the juxtaposition of Paleozoic and Precambrian rocks across the plateau marginal fault beneath the Peach Springs Tuff.

The physiography along the edge of the plateau may have consisted of low cuestas breached by northeast-flowing, consequent trunk streams with subsequent tributaries paralleling the more resistant strata. West-facing alluvial fans could have formed between the major streams. Such a drainage system would explain the trends of buried canyons through Milkweed, Hindu, and Peach Springs Canyons.

The Hurricane fault obviously influenced development of the wider valley through Peach Springs Canyon and the Truxton Valley. The volcanic and sedimentary rocks which later filled the old channels were deposited before the development of the modern Colorado River and its tributaries on the Hualapai Plateau. Remnants of high-level gravel capped by thin basalt flows on the south rim of the Grand Canyon at the head of Separation Canyon could not have been carried across the modern Colorado River tributaries which now surround the outcrop area. Pre-tuff gravel outcrops north

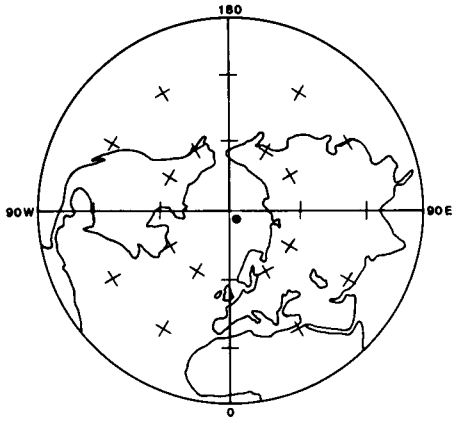


Figure 4. Approximation of middle Miocene paleomagnetic pole for northwestern Arizona. Long 34.4° E., lat 84.1° N., δp = 8.4°, δm = 12.7°.

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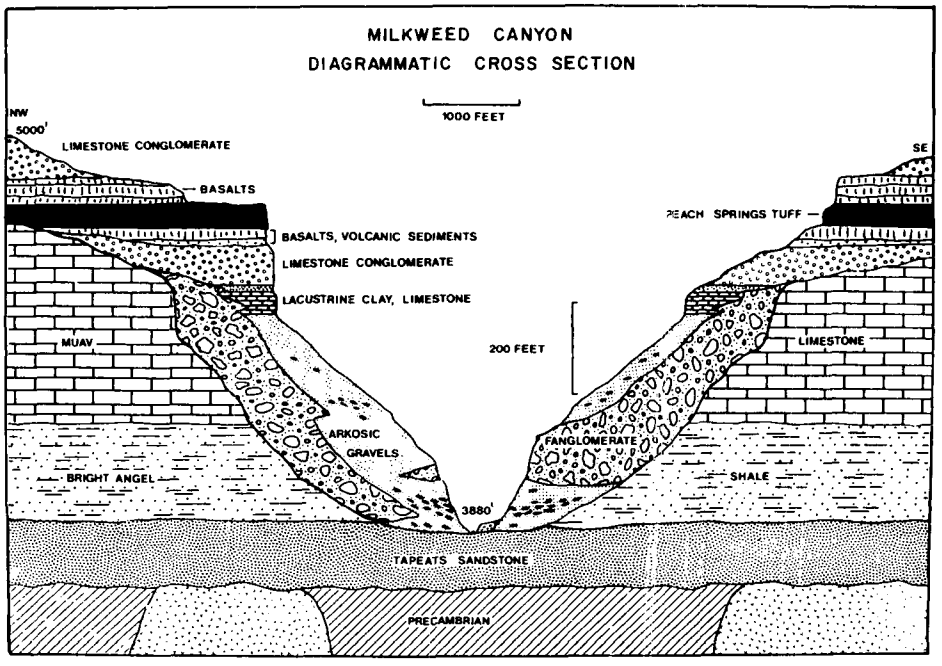


Figure 5. Composite cross-section view of Cenozoic deposits in Milkweed Canyon ~5 mi from edge of plateau.

of Hindu Canyon indicate that alluvial fans(?) containing fragments of Coconino Sandstone had extended southwestward across the present location of the modern Colorado River from the Shivwits Plateau (Fig. 1). Coconino Sandstone is not present on the Hualapai Plateau, and north of the Colorado River it is stratigraphically 1,000 ft above the gravel outcrops near Hindu Canyon.

A postulated regional drainage northward through the Shivwits Plateau (along the Hurricane fault zone) from the channel incised in Peach Springs Canyon would require more regional northeasterly tilting (uplift to the west and southwest) of the entire edge of the plateau than the Paleozoic rocks have today. Imbrication measurements show that the buried channel, with its basal arkosic gravel in Milkweed Canyon (Fig. 5), connected with Hindu Canyon and then with lower Peach Springs Canyon (Hurricane fault zone). Similar arkosic gravels of younger but comparable ages on the Coconino Plateau east of the Hurricane fault were carried northward from central Arizona and possibly may have crossed the position of the modern Colorado River at a higher elevation (Young, 1970; McKee and others, 1967, p. 55–56; McKee and McKee, 1972). (I. Lucchitta is studying similar arkosic gravels beneath Tertiary volcanic rocks on the Shivwits Plateau [1973, oral and written communs.].) Greater northeast tilting (~1°) of the region along the west and south margins of the plateau, followed by block faulting and relative subsidence of the Great Basin, could explain development of the drainage west of the Hurricane fault on the Hualapai Plateau and its subsequent disruption. Such a sequence of events is supported by outcrop elevations of the Peach Springs Tuff from the west edge of the plateau westward to Kingman. Tuff outcrops near Kingman (closer to the source) average 1,000 ft below the tuff outcrops along the edge of the plateau, demonstrating relative regional collapse of the eastern edge of the Great Basin or uplift of the plateau some time after eruption of the tuff. In the Cerbat Mountains, however, Precambrian rocks are still at higher elevations than in the areas to the north and on the plateau.

The generally synchronous nature of block faulting and volcanism northwest of the study area, near Lake Mead, has been documented by Anderson and others (1972), McKee and Anderson (1971), and McKee (1971).

The Peach Springs Tuff is also displaced by the major portion of movement along the Hurricane fault in upper Peach Springs Canyon (200 ft) and the Truxton Valley (100 ft or more).

The Hurricane fault intersects the Cottonwood Cliffs where a ½-mi-wide shear zone exposes vertically dipping beds of Precambrian metaconglomerate.

Of significance to the erosional and drainage history of the region is the width (15 mi) of the Truxton Valley, a large re-entrant in the edge of the plateau; this width indicates a concentrated and effective removal of Paleozoic rocks, begun by drainage along the Hurricane fault zone, to a degree not found in similarly located drainages to the north or south. No other valleys cut through the edge of the plateau are as wide, and only the Colorado Canyon is as deep. This erosion probably indicates early (pre–middle Miocene) movement along the Hurricane fault to allow considerable drainage development and valley widening in the Paleozoic rocks along the fault zone before deposition of the prevolcanic gravels and before filling of the entire valley by the Peach Springs Tuff.

IMPORTANT REGIONAL CORRELATIONS

Two other significant volcanic centers in the study area that are closely related in time to the Peach Springs Tuff are the Aquarius Mountains caldera and the Mohon Mountains volcanic field on the north and south sides of Trout Creek, respectively (Fig. 1). The Peach Springs Tuff extends as far south as the drainage of Trout Creek, and there it overlies the lapilli tuffs of the Aquarius Mountains and disappears beneath the basalts of the Mohon Mountains. The tuff wedges out in the canyon walls of Trout Creek. Anderson and others' (1955) detailed map and report on the south side of the Mohon Mountains near Bagdad includes no mention of a similar, thin ash flow beneath the lava that Anderson named the Sanders Basalt; this lava represents the southern extension of the Mohon Mountain flows.

The Aquarius Mountains caldera is a large, circular, volcanic complex with 1,500 ft of relief, ~11 mi wide at its base, with a central, eroded crater ~3 mi across. Most of the surrounding deposits are lapilli tuffs interspaced with thinner flow breccia and lava, which contain large blocks of the grayish andesitic(?) porphyry that crops out in the central crater. Basaltic or andesitic flows throughout the Aquarius Mountains section appear to be thin and discontinuous. The lapilli tuffs of the Aquarius Mountains, which show some faulting but have not been extensively deformed, unconformably overlie basaltic(?) flows and cinder cones erupted onto an eroded Precambrian surface near the mouth of Trout Creek. This section, north of the Mohon Mountains, generally resembles the sequence of deposits in the Bagdad area described by Anderson and others (1955). Some of the pumice and lapilli tuff in the sediments Anderson described might have originated in the Aquarius Mountains. If so, some of the tentative age assignments made by Anderson should be increased.

The Pliocene lake beds west of the Mohon Mountains in the Big Sandy Valley (Sheppard and Gude, 1972) probably contain tuffs representing the volcanism associated with the growth of the Mohon Mountains volcanic pile.

The Peach Springs Tuff flowed against the

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western flank of the Aquarius Mountains caldera ~800 ft below its present summit and occurs on small buttes, which cap thick sections of lapilli tuffs. The tuff was covered by Mohon Mountains basalt, and excellent exposures of the relationships of all the volcanic units can be seen in the sides of Trout Creek Canyon. The observation of field relations between the Peach Springs Tuff and these other volcanic centers might eventually enable correlation of regional Cenozoic events from Kingman eastward to the Chino Valley–Prescott area. This is an area where lack of detailed correlations has prevented precise reconstruction of the Cenozoic history of the evolution of the Basin and Range–Colorado Plateau boundary.

EROSION OF THE WESTERN GRAND CANYON

The volcanic rocks in many places on the edge of the plateau are overlain by gravel derived from areas of local high relief such as Music Mountain, the sides of the Truxton Valley, and escarpments to the east of the Cottonwood Cliffs and Aquarius Mountains. On the Hualapai Plateau, the Colorado River tributaries are incised in these postvolcanic gravel deposits and all older rocks. This postvolcanic gravel accumulated in the Truxton Valley until Pleistocene time (Twenter, 1962, p. 21). The Truxton Valley, however, was an enclosed basin, only recently affected by headward erosion of drainage from the west, whereas the Colorado River tributary drainage must have become well established somewhat earlier, to the north on the Hualapai Plateau.

The relative youth of the western Grand Canyon drainage is further substantiated on the Hualapai Plateau, where Bridge Canyon, a north-flowing Colorado tributary, has diverted (captured) tributaries which formerly flowed south into the prevolcanic channel through Hindu Canyon (Young, 1970).

CONCLUSIONS

The eruption of the Peach Springs Tuff appears to have accompanied or immediately preceded most of the volcanism and block faulting (middle Miocene to early Pliocene) in Mohave County. After this time the Colorado Plateau became more obviously structurally separated from the Great Basin, as indicated by major drainage changes and diversions and by fault displacement observable in the tuff, on the Hurricane fault in Peach Springs Canyon, and along the Cottonwood Cliffs. Pre–Colorado River drainage flowing north through Peach Springs Canyon along the Hurricane fault may have been started by greater regional structural tilting to the northeast associated with major uplift adjacent to the plateau in pre–middle Miocene time. This drainage was terminated by the events associated with the formation of the Basin and Range structures west of the plateau. Prior to the post-tuff structural differentiation of these two provinces, the edge of the Colorado Plateau from Lake Mead south to the Mohon Mountains appears to have been a structural boundary which had undoubtedly experienced some minor faulting during the regional upwarping (greater to the west). Evidence for this assumption is (1) the abrupt juxtaposition of Precambrian and Paleozoic rocks beneath the volcanic rocks along the edge of the Colorado Plateau; (2) minor, local tributary drainage flowing westward off the edge of the plateau into the Big Sandy Valley (in pre-tuff time); and (3) stratigraphic relations in the Aquarius Mountains, where pre–Peach Springs Tuff volcanic rocks of two different eruptive episodes are separated by an unconformity and rest on eroded Precambrian rocks at the margin of the plateau.

The Peach Springs Tuff blocked all previously existing drainage along the margin of the plateau from the present Colorado River southward to Trout Creek in middle Miocene time. Faulting which followed the eruption of the tuff resulted in the relative, regional subsidence of the area from Kingman to the edge of the plateau by a minimum of 1,000 ft, as recorded by the distribution and elevations of the tuff outcrops (Fig. 2). Following (or accompanying) the faulting which produced this regional elevation difference, interior basin deposits with extensive lacustrine facies, such as the Muddy Creek deposits of Miocene to Pliocene age near Lake Mead and the Big Sandy Formation of Pliocene age west of the Mohon Mountains, were common west of the plateau. All of these events indicate a middle to late Pliocene age for the development of the Colorado River to the west of the Hurricane fault and across the Hualapai Plateau.

McKee and others (1967) have suggested that the modern Colorado River developed on the Hualapai Plateau by headward erosion from the west and by diversion of remnants of the older Hualapai drainage system. This might have been followed by headward erosion of the drainage along the Hurricane fault zone and eastward to capture, eventually, drainage around the southern end of the Kaibab upwarp. The former outlet for the drainage east of the Kaibab upwarp has not been absolutely determined (see summary in Hunt, 1969). It is possible that in Miocene or early Pliocene time drainage from east of the Kaibab upwarp could also have continued around the southern end of the uplift and drained northward, joining the drainage from the Peach Springs area. However, any complete explanation of the regional drainage development must account for the removal of the Mesozoic rocks above the Kaibab Limestone from most of northern Arizona west of the Little Colorado River *before* the development of the modern Grand Canyon drainage system in early to middle Pliocene time.

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